

Science — Cbse Periodic Test (Pt) — 40 Marks

Subject: Science | Class: Class 9

Total Marks: 40 | Time Allowed: 1h 30m | CBSE

General Instructions:

1. Read all questions carefully before answering.
2. All questions are compulsory unless stated otherwise.
3. Marks for each question are indicated on the right.
4. Write neatly and legibly.

Q.NO.	QUESTIONS	MARKS
Section A Attempt all 8 questions. Each carries 1 mark(s).		
1	An animal is observed in a garden. Which feature confirms that it is an insect rather than an earthworm? (A) Bilateral symmetrical body (B) Cylindrical body (C) Body with little segmentation (D) Body with jointed legs <i>[Ch 12: Patterns in Life: Diversity and Classification]</i>	1
2	What kind of elements generally donate their valence electrons to achieve a stable electronic configuration? (A) Elements with exactly four valence electrons (B) Elements with less than four valence electrons in their valence shell (C) Elements with more than four valence electrons (D) Elements with eight valence electrons <i>[Ch 9: Atomic Foundations of Matter]</i>	1
3	A cart starts from rest with zero velocity and travels a distance s with acceleration a in time T. Which kinematic equation correctly relates these variables? (A) $s = aT$ (B) $s = 1/2aT^2$ (C) $s = 1/2aT$ (D) $s = aT^2$ <i>[Ch 6: How Forces Affect Motion]</i>	1



Q.NO.	QUESTIONS	MARKS
4	Flowers that bloom at night are generally white or light in colour compared to flowers that bloom during the day. What is the most likely reason for this? (A) Light colours help the flowers retain more water at night (B) Dark colours would cause the flowers to overheat during the night (C) Light colours make the flowers more visible to pollinators that are active at night (D) Light colours prevent the flowers from closing their petals at night <i>[Ch 11: Reproduction: How Life Continues]</i>	1
5	What is the process by which meristematic tissue becomes specialised to perform specific functions called? (A) Differentiation (B) Transpiration (C) Diffusion (D) Conduction <i>[Ch 3: Tissues in Action]</i>	1
6	Assertion (A): Sclerenchyma tissue provides hardness and strength to plants like the coconut husk and walnut shell. Reason (R): Sclerenchyma cells have thick walls due to the deposition of pectin. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 3: Tissues in Action]</i>	1
7	Assertion (A): Work is done by a force on an object when the force displaces the object in the direction of the force. Reason (R): An object having the capacity to do work is said to possess energy. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 7: Work, Energy, and Simple Machines]</i>	1
8	Which pair of cell organelles contains DNA? (A) Mitochondria and Nucleus (B) Ribosomes and Lysosomes (C) Golgi bodies and Ribosomes (D) Nucleus and Lysosomes <i>[Ch 2: Cell: The Building Block of Life]</i>	1
Section B Attempt all 4 questions. Each carries 2 mark(s).		



Q.NO.	QUESTIONS	MARKS
9	How can one scientifically disprove the claim that food becomes harmful during an eclipse? <i>[Ch 1: Exploration: Entering the World of Secondary Science]</i>	2
10	If 25 cells are seen along the diameter of a visible field measuring 5000 μm , calculate the estimated size of one cell. <i>[Ch 2: Cell: The Building Block of Life]</i>	2
11	Fill in the blanks. Among magnesium and chlorine, magnesium atom can give two electrons to become Mg^{2+} . However, chlorine can take only one electron to become _____. Now, _____ ion of magnesium and _____ ions of chlorine combine to give magnesium chloride. <i>[Ch 9: Atomic Foundations of Matter]</i>	2
12	Name the four components of xylem. Which one is the only living component? <i>[Ch 3: Tissues in Action]</i>	2
Section C Attempt 3 out of 4.		
13	Describe the structure of the cell membrane as explained by the fluid-mosaic model. <i>[Ch 2: Cell: The Building Block of Life]</i>	3
	OR Many smartphones have an inbuilt accelerometer that can detect very small accelerations. Install an app, such as Phyphox (phyphox.org) and open 'Accelerometer' (without g). Note the readings when (i) the phone is on an outstretched palm, and (ii) the phone is kept on the floor. What differences do you observe? What does this tell you about motion and acceleration in real situations? (a) What differences do you observe between the readings when the phone is on an outstretched palm and when kept on the floor? <i>[Ch 4: Describing Motion Around Us]</i>	
14	Explain the role of prediction in science. What do scientists do when their predictions do not match their observations? <i>[Ch 1: Exploration: Entering the World of Secondary Science]</i>	3
15	F. C. Steward grew phloem cells of a carrot in a nutrient medium with simple sugars and hormones, observing the formation of a complete plant. Explain the processes of dedifferentiation and redifferentiation that occurred, and state the term for this cellular ability. <i>[Ch 3: Tissues in Action]</i>	3
Section D Attempt 2 out of 3.		



Q.NO.	QUESTIONS	MARKS
16	Differentiate between laws, theories, and principles in science. Explain each term with an example and discuss why a theory in science is not the same as a guess. <i>[Ch 1: Exploration: Entering the World of Secondary Science]</i>	5
	OR Explain how a microscope enables us to study cell structure. Describe the three main features that scientists have improved in microscopes over the years, and demonstrate how the estimated size of a cell and its total magnification are calculated. <i>[Ch 2: Cell: The Building Block of Life]</i>	
17	Distinguish between voluntary and involuntary movements. Compare the structural features and functions of skeletal muscles, smooth muscles, and cardiac muscles. <i>[Ch 3: Tissues in Action]</i>	5
Section E Attempt 1 out of 1.		
18	Case Study: Building Scientific Models A student wants to model how long it takes for a ball thrown upward to return to the ground. She considers several factors: the mass of the ball, its initial speed and direction, air resistance, the colour of the ball, and the spin of the ball. She recalls that physicist Meghnad Saha simplified the study of stars by ignoring complex processes and focusing only on temperature, pressure, and how atoms formed ions, which helped explain the connection between star colour and temperature. She decides to apply a similar simplification approach. She also notes that science uses precise symbols like m for mass, v for velocity, and F for force, and that equations are compact statements about relationships between quantities. (i) What three factors did Meghnad Saha focus on to explain the colour of stars? (ii) Why should the student ignore the colour of the ball when modeling the time it takes to return to the ground? (iii) The student ignores air resistance and spin in her simple model. Later, she wants a more accurate prediction. What should she do, based on the approach described for building complex models? (iv) The student writes an equation relating distance, time, and velocity to describe the ball's motion. Explain why this equation is more than just a calculation tool. <i>[Ch 1: Exploration: Entering the World of Secondary Science]</i>	5

